Supplementary Information for:

Diffractive cascades for polychromatic hard X-ray focusing

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S1. OPTIMIZATION DETAILS

In this section we specify the mathematical details of the optimization procedure. We also provide the physical and computational parameters used in the experiments presented in the main text. We first define the optimization problem in question as

$$\min_{\boldsymbol{\rho}} \mathcal{L}(\boldsymbol{\rho}) \tag{S1}$$

$$\text{subject to: } g_{\text{solid}}(\bar{\boldsymbol{\rho}}) \leq 0 \tag{S2}$$

$$g_{\text{void}}(\bar{\boldsymbol{\rho}}) \leq 0 \tag{S3}$$

$$0 \leq \rho_l(r) \leq 1 \quad \forall l, r \tag{S4}$$

Where $\boldsymbol{\rho} = \{\rho_l(r)\}_{l=1}^{N}$ is the set of radial density profiles for each of the N elements. The forward model consists of interleaved propagation through free space and the application of the cascade elements. The application of an element transforms the incoming wave according to the thin element approximation

$$U_l^+(r,\lambda) = U_{l-1}^-(r,\lambda) \cdot t(\bar{\rho}_l(r),\lambda), \tag{S5}$$

where $t(\bar{\rho}_l(r),\lambda)$ is the complex transmission function of the l-th element at the wavelength λ. The wavefront incident on the next element is determined using the angular spectrum propagation kernel

$$H(f_x,f_y) = \exp\left[i2\pi\frac{z}{\lambda}\sqrt{1-(\lambda f_x)^2-(\lambda f_y)^2}\right]. \tag{S6}$$

With this the expression for the field after propagation is

$$U_l^-(r,\lambda) = \mathcal{F}^{-1}\{H(f_x,f_y)\cdot\mathcal{F}\{U_l^+(r,\lambda)\}\}, \tag{S7}$$

where $\mathcal{F}\{\cdot\}$ is the Fourier transform. The objective function is computed as the integrated power within a small region Ω around the center of the focal plane

$$\mathcal{L}(\boldsymbol{\rho}) = -\sum_{\lambda}\int_{\Omega}|U_N^-(r,\lambda)|^2. \tag{S8}$$

The forward model is evaluated on the parameters after they have been filtered and projected. The filter function filters out features below a given size. It is defined as,

$$f_{\text{filter}}(\rho_l(r)) = \tilde{\rho}_l(r) = \int K(r-r')\rho_l(r')dr', \tag{S9}$$

where $K(r)$ is the spatial kernel

$$K(r) = \frac{w(r)}{\int w(r')dr'}. \tag{S10}$$

$w(r) = \max(0, R - |r|)$ is the weighting function with R being the desired minimum feature size. $\bar{\rho}_l(r)$ is obtained by applying a threshold function to the filtered parameters

$$\bar{\rho}_l(r) = f_{\text{thresh}}(\tilde{\rho}_l(r); \beta) = \frac{\tanh(\beta/2) + \tanh[\beta(\tilde{\rho}_l(r) - 1/2)]}{2\tanh(\beta/2)}, \tag{S11}$$

where β controls the strength of the projection. To further ensure that no features below the minimum are included in the final design, we supply inequality constraints to the optimization algorithm. These constraint functions are computed for each element and then aggregated together. For a single element the feature size constraints are computed as

$$g_{s,l} = |\max(0, \bar{\rho}_l(r) - \mathcal{O}(\bar{\rho}_l(r))| - \epsilon \tag{S12}$$

$$g_{v,l} = |\max(0, \mathcal{C}(\bar{\rho}_l(r)) - \bar{\rho}_l(r)| - \epsilon \tag{S13}$$

Where ϵ is the constraint violation tolerance. $\mathcal{O}(\cdot)$ and $\mathcal{C}(\cdot)$ are the smooth opening and smooth closing operations as defined in [1]. These per-element constraint functions are then aggregated into a single number for the solid and void regions respectively

$$g_{\text{solid}}(\bar{\boldsymbol{\rho}}) = \frac{1}{\beta_{\text{agg}}} \ln\left(\sum_{l=1}^{N} \exp(\beta_{\text{agg}} \cdot g_{s,l})\right), \tag{S14}$$

$$g_{\text{void}}(\bar{\boldsymbol{\rho}}) = \frac{1}{\beta_{\text{agg}}} \ln\left(\sum_{l=1}^{N} \exp(\beta_{\text{agg}} \cdot g_{v,l})\right). \tag{S15}$$

Below we list the parameters used for the forward model and the optimization procedure. These parameters are used for all the results in the main text except where otherwise noted. For example, in the plots where we show cascade performance as a function of source bandwidth and energy, these parameters vary while all other parameters take the values in the tables below. The energy-dependent refractive indices of the materials considered in this work are taken from [2].

Parameter	**Value**
Central X-ray energy	10 keV
Source energy bandwidth	10^{-4}
Minimum feature size	50 nm
Number of elements	10
Source wavelengths	11
Grid spacing	10 nm
Element diameter	82 μm
Zone thickness	400 nm
Membrane thickness	1 μm
Inter-element distance	1 cm
Focal length	33 mm

TABLE S1: Physical parameters.

Parameter	**Value**
Maximum iterations	100
β_{min}	1
β_{max}	256
β_{agg}	10
ϵ	10^{-6}
Ω diameter	24 nm

TABLE S2: Optimization parameters.

S2. EFFECT OF PARTIAL SPATIAL COHERENCE

While modern synchrotron and free-electron laser sources are highly coherent, full spatial coherence is still an imperfect assumption. In this section we introduce a method for modeling partial spatial coherence and apply it to diffractive cascades.

Previous studies of partially coherent X-ray radiation have utilized Monte Carlo simulations [3]. However, a more computationally elegant method uses the cross spectral density. The cross spectral density quantifies the correlation between pairs of points on a wavefront [4]. A standard form of the cross spectral density is the Gaussian-Schell model which we write here in polar coordinates

$$W(r_1, \phi_1, r_2, \phi_2) = \exp\left[-\left(\frac{1}{4\sigma_s^2} + \frac{1}{2\sigma_g^2}\right)(r_1^2 + r_2^2)\right] \exp\left[\frac{r_1 r_2 \cos(\phi_1 - \phi_2)}{\sigma_g^2}\right], \tag{S16}$$

where σ_s is the beam width and σ_g is the coherence length. For efficient computation it is convenient to expand the cross spectral density into coherent modes

$$W(r_1, \phi_1, r_2, \phi_2) = \sum_{l=-\infty}^{\infty} \sum_{p=0}^{\infty} w_{lp} \Psi_{lp}(r_1, \phi_1) \Psi_{lp}^*(r_2, \phi_2), \tag{S17}$$

where $\Psi_{lp}(r, \phi)$ are the Laguerre-Gaussian modes and w_{lp} their corresponding weights. Since the modes are mutually incoherent we can propagate each of them individually and combine their results at the end. To determine the effect of partial coherence on the performance of our diffractive cascades, we replaced the fully spatially coherent illumination used in the main text with a Gaussian-Schell model beam. In particular, for each combination of σ_s and σ_g we computed the coherent mode decomposition of the Gaussian-Schell model. We then truncated the decomposition where w_{lp} dropped below 10^{-3}. This ensures that the decomposition faithfully represents the true coherence properties of the beam. Figure S1 shows the focusing efficiency of diffractive cascades designed for different levels of spatial coherence. The fully coherent limit is in the top right of the plot and corresponds to the limits $\sigma_s/D, \sigma_g/D \gg 1$, where D is the diameter of a cascade element. We observe that diffractive cascades maintain their performance even for modest spatial coherence of the beam.

The efficient performance of diffractive cascades under partially coherent illumination suggests they are effective with multiple common X-ray sources. The most coherent X-ray sources are free-electron lasers, which often have spatial coherence lengths exceeding their beam size, resulting is essentially full spatial coherence [5]. Slightly less coherent are synchrotron sources which can

achieve coherence lengths of 100-200 μm [6]. With further optimization for partial coherence, laboratory-scale sources could also potentially be used with diffractive cascades. Liquid metal jets have small spot sizes and are capable of coherence lengths of roughly 10 μm [7], just beyond the point where efficiency drops off steeply in Figure S1.

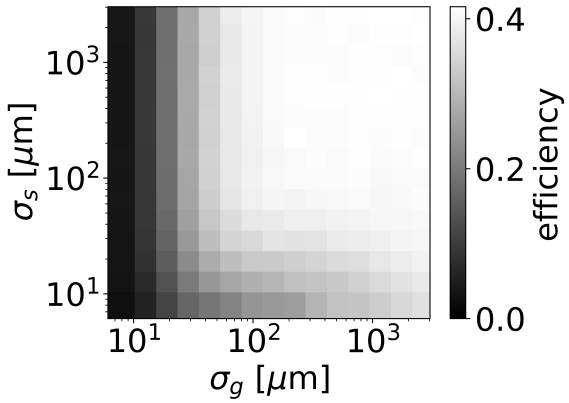


FIG. S1: **Effect of partial coherence on cascade performance.** Plot of the efficiency of cascades designed for different degrees of partial spatial coherence. For each combination of σ_s and σ_g a cascade is designed with the corresponding partially coherent beam in the forward model. The physical and optimization parameters are those specified in Tables S1 and S2. For the fully coherent model, the cascade with these parameters achieved a focusing efficiency of 41%.

S3. PHYSICAL FACTORS GOVERNING THE OPTIMAL NUMBER OF CASCADE ELEMENTS

In this section we analyze the physical factors that determine the optimal number of elements in a diffractive cascade. The empirical curves in Figure 1(e) show that cascade performance peaks at a material-dependent value of N, then declines. This behavior reflects a competition between two opposing effects: the increased focusing ability afforded by additional diffractive elements, and the increased absorption incurred by the X-rays passing through those elements. We construct a simple model to illustrate how these two effects interact.

We write the figure of merit as a product of two terms,

$$FOM(N) = T(N)\eta(N). \tag{S18}$$

$T(N)$ is the transmission of a cascade with N elements, quantifying the loss due to absorption as the X-rays pass through additional cascade elements. $\eta(N)$captures the saturation of focusing performance with increasing N: adding elements expands the space of possible designs and improves performance, but with diminishing returns, since each additional element contributes less incremental benefit when the cascade already contains many elements.

Computing $T(N)$ is simply a matter of considering the absorption due to multiple layers of material with a given refractive index. We do not consider reflection between air and the membrane or the membrane and the element material due to the negligible refractive index contrast. The transmission as a function of the number of cascade elements is

$$T(N) = T_1^N = \exp\left[-N\frac{4\pi}{\lambda}\left(\beta_m t_m + f\beta_z t_z\right)\right], \tag{S19}$$

where $n_m = 1-\delta_m+i\beta_m$ and t_m are the refractive index and thickness of the membrane material, respectively. $n_z = 1-\delta_z+i\beta_z$ and t_z are the refractive index and thickness of the element zones. f is the average fill factor of the elements in the cascade. Because $T(N)$ depends directly on the imaginary part of the refractive index of the zone material, it differs substantially across materials: heavier elements such as gold are far more absorptive than lighter ones such as silicon, causing $T(N)$ to decay more rapidly with N. This explains qualitatively why the optimal number of elements is smaller for denser, more absorptive materials.

The saturation of focusing performance with N is modeled as

$$\eta(N) = 1 - e^{-kN}, \tag{S20}$$

where k is a rate parameter controlling how quickly performance saturates as elements are added. This functional form reflects the expectation that the marginal benefit of an additional element decreases as the cascade grows. The product $T(N)\eta(N)$ therefore rises initially, as the gain from additional elements outweighs the cost of absorption, and then falls once absorption becomes dominant.

In Figure S2 we plot this figure of merit as a function of N for different materials, using the fill factor at each point taken from the corresponding design in Figure 1(e). The model follows a similar trend to the one found empirically, and correctly captures the ordering of materials by their optimal N. This agreement indicates that the competition between absorption and the saturation of focusing performance is the principal physical mechanism governing the optimal cascade depth, and that material choice enters primarily through its effect on $T(N)$.

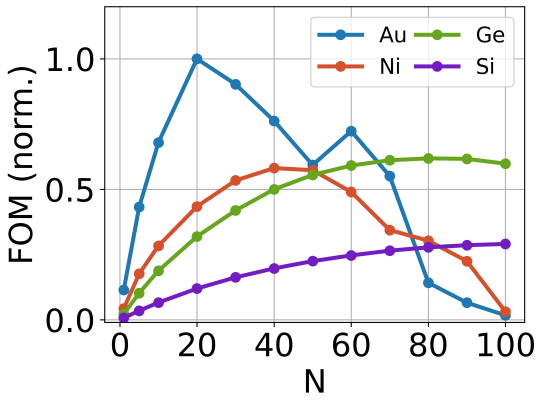


FIG. S2: **Cascade element tradeoff figure of merit.** Plot of the figure of merit as a function of the number of elements in the cascade for different zone materials. For each combination of a material and N, the fill factor is determined by the corresponding optimization in Figure 1(e). The learning rate for each material is determined by fitting the k parameter in Eqn. S20 based on the first three point of the efficiency plot in Figure 1(e).

S4. CASCADE PERFORMANCE AS A FUNCTION OF FOCAL LENGTH AND INTER-ELEMENT DISTANCE

One of the advantages of diffractive cascades is their ability to accommodate a range of focal lengths. In order to change the focal length of a zone plate one must change the zone plate diameter or feature size. Changing the focal length of a diffractive cascade is only a matter of moving the objective region to a different plane.

Similarly, diffractive cascades can accommodate a range of distances between the elements. While efficiency peaks when the distance between elements is roughly 5 mm, efficiency is high across a range of distances. This allows diffractive cascades to conform to the constraints of the system they are integrated into.

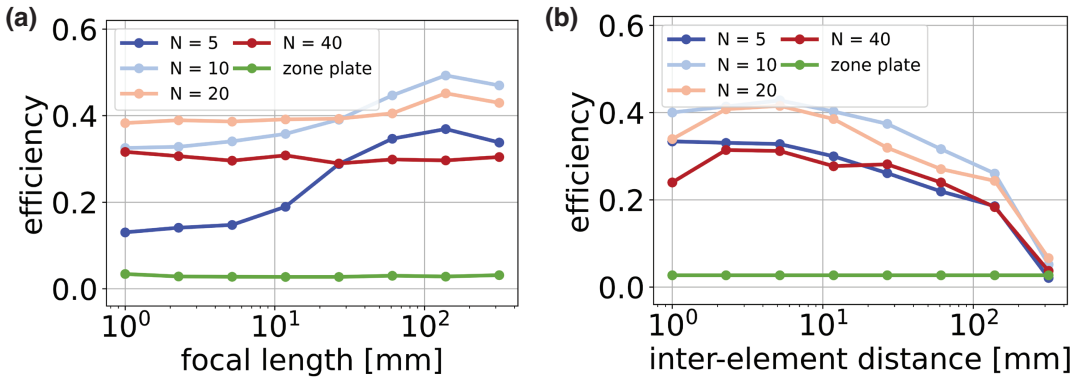


FIG. S3: **Cascade performance as a function of focal length and inter-element distance. (a)** For each focal length a cascade is designed using the specified plane in the objective function. The zone plate focal length is varied by changing the minimum feature size to keep the element diameter constant. **(b)** For each inter-element distance a distinct cascade is designed. The zone plate design is identical for all values of the inter-element spacing.

S5. ROBUSTNESS OF CASCADES TO DILATION, TRANSITION SMOOTHNESS, AND TILT

In the main text, we showed the robustness of diffractive cascades to erosion, or over-etching. The complementary phenomena is dilation, or under-etching. Another possible fabrication error is smearing of the transitions between solid and void regions. Beyond these, another alignment error that can occur is from the tilt of elements relative to their nominal angles perpendicular to the beam. The effect of all three of these imperfections is shown in Figure S4.

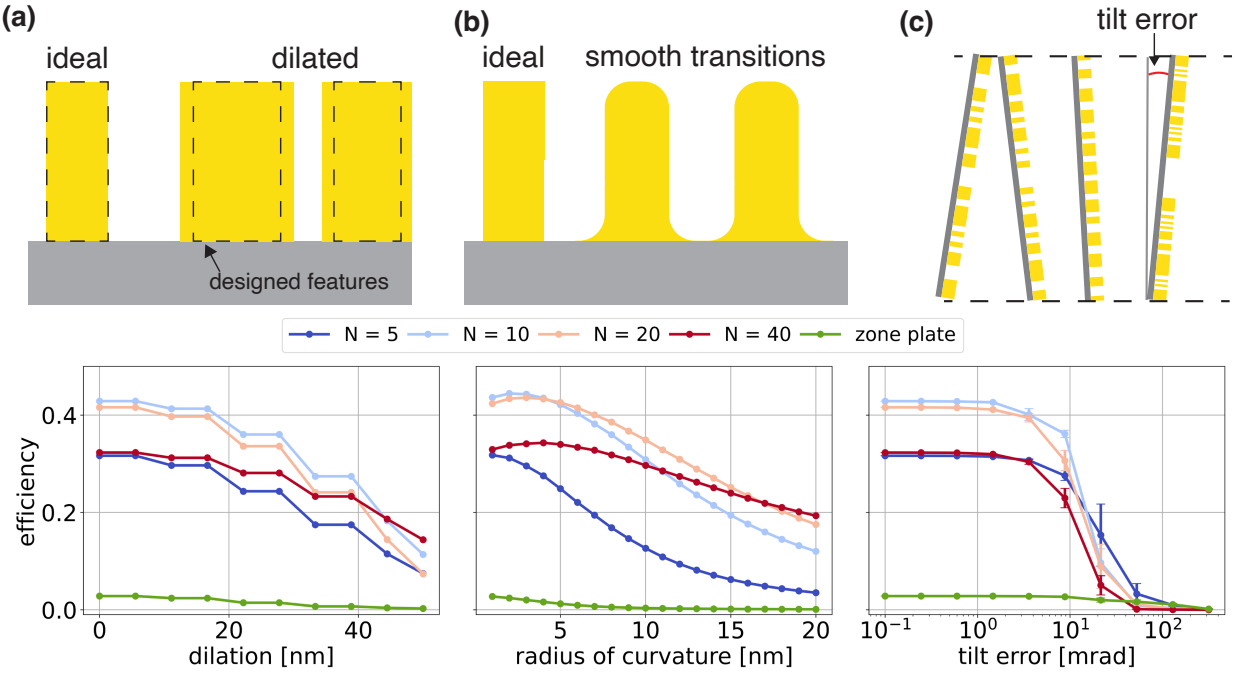


FIG. S4: **Robustness of cascades and zone plates to dilation and sidewall smearing.** **(a)** Effect of dilation or under-etching on cascade and zone plate performance. Dilation refers to the amount by which each feature is expanded. **(b)** Effect of smooth solid-void transitions on cascade and zone plate performance. The radius of curvature refers to the radius of the corners of each zone. Zero radius of curvature corresponds to a perfectly abrupt transition. **(c)** Effect of tilt error on cascade and zone plate performance. For each level of tilt error, the elements of the cascade are randomly tilted in the yaw and pitch directions according to a uniform distribution whose mean is the given tilt error level. For each error level multiple trials are conducted with different random angles to generate the error bars in the plot.

We observe that the cascades are similarly robust to dilation as they are erosion. For cascades with large numbers of elements, some dilation is actually moderately beneficial to performance.

Similarly, for some cascades, small amounts of sidewall smearing increase the performance of the system before causing degradation. We conclude that diffractive cascades outperform zone plates even for large amounts of dilation and transition smoothening. Similarly, in the tilt robustness study we introduced random tilts to the elements in the cascade by randomly changing the pitch and yaw of the elements. The plot in Figure S4 plots the efficiency of different cascades against the mean tilt error imposed. We observe that tilt errors below 1 milliradian are tolerable, which is well within the capability of modern piezoelectric stages used in beamlines.

S6. COMPARISON TO FRESNEL ZONE PLATE CASCADE

In the main text we made comparisons between the optimized diffractive cascades and a single Fresnel zone plate. Cascades of zone plates have also been used to improve focusing ability by increasing the effective thickness of the focusing optic [8–12]. However, there are several issues with this approach. One issue is the fact that the spacing between zone plates in a cascade must be small. Initial analysis imposed a strict near-field spacing between the zone plates [13], but it was later shown that additional zone plates in the intermediate-field also improves focusing, allowing inter-element spacings on the millimeter scale [9]. Nevertheless, even with the relaxed constraint, performance quickly degrades as more elements are cascaded. Separately, zone plate cascades also inherit the monochromatic nature of zone plates. This means they cannot focus large bandwidth sources like inverse designed diffractive cascades can.

In this section we compare the performance of our optimized cascades to cascades of zone plates, whose radii have been altered to allow their foci to coincide. In Figure S5(a) we show a performance comparison between a zone plate cascade, an optimized cascade, and a single zone plate. We observe that for low element numbers, the zone plate and optimized cascades have similar performance. However, as a the element number increases, the optimized cascade reaches a higher efficiency. In Figure S5(b), we compare the optimized and zone plate cascade performance as a function of the bandwidth and energy of the source, as well as the aspect ratio of the individual elements. We observe that utilizing optimization allows more of the space of source parameters to be focused at higher efficiencies. Our optimized diffractive cascades can maintain high efficiency deep into the hard X-ray regime where low aspect ratio zone plate cascades begin to fail. Since zone plate stacking is most useful at small element separations, the plots below were generated for an inter-element distance of 1 mm, smaller than the optimal of 1 cm for our optimized cascades.

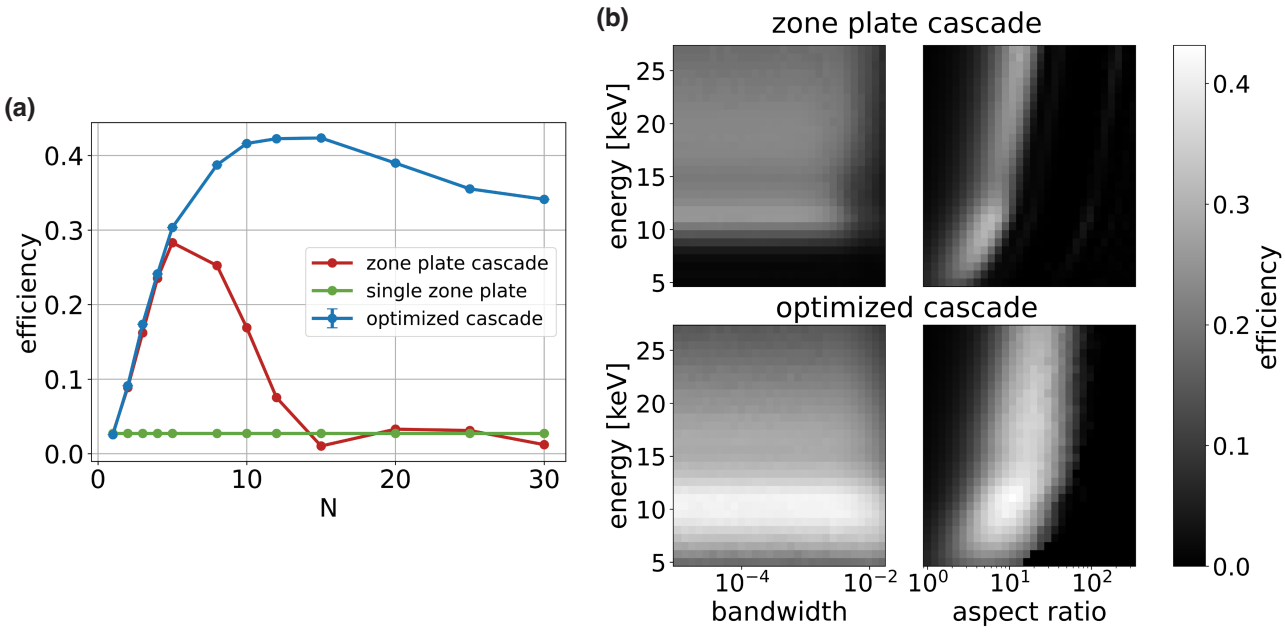


FIG. S5: **Zone plate and optimized cascade comparison.** **(a)** Zone plate cascade, optimized diffractive cascade, and single zone plate focusing performance as a function of number of elements. Parameters used are the standard ones as noted in Tables S1 and S2, except for the inter-element spacing, which was reduced to maximize zone plate cascade performance. **(b)** Zone plate and optimized cascade performance as a function of source energy and bandwidth, as well as element aspect ratio.

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